

EduAlert: Early Identification of At-Risk Students Using Data Mining

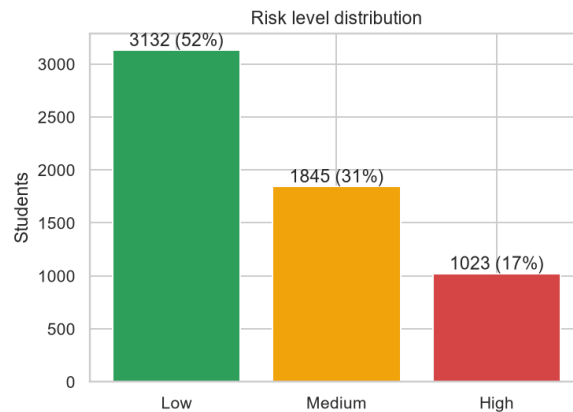
Model report

1. Problem and approach

Struggling students are usually identified only after final results. EduAlert uses earlier-period academic, attendance, assessment and engagement data to flag students at Low / Medium / High risk early enough for support to help, and explains each flag with its main warning signals.

2. Data

Synthetic, anonymised data (not real student records): 6,000 students, 20 model features. Risk_Level is derived from a *later* outcome (final score: High <50, Medium 50-65, Low ≥65); the final score itself is never a model feature, which avoids target leakage. Class balance: Low 3132 (52%), Medium 1845 (31%), High 1023 (17%).



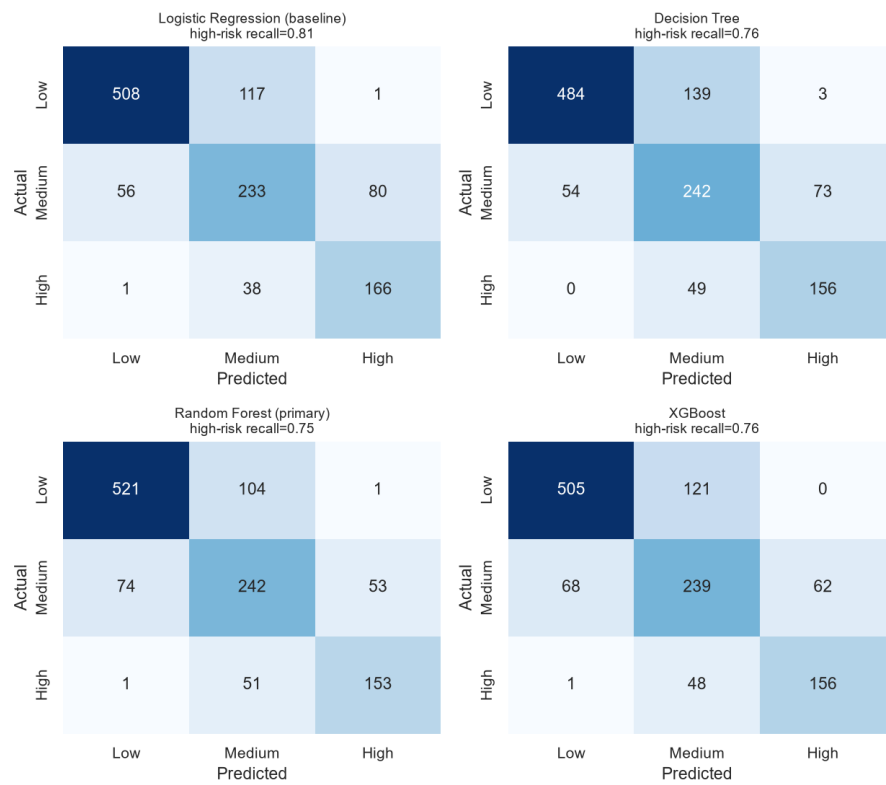
3. Pipeline

Remove duplicates, null impossible values, median/mode imputation, outlier capping, feature engineering (Assessment_Avg, Attendance_Status, Engagement_Score, Academic_Performance_Index), one-hot encoding, standard scaling, stratified 80/20 split, Random Forest tuned by 3-fold grid search (best: {'max_depth': 12, 'min_samples_leaf': 3, 'n_estimators': 200}).

4. Results (held-out test set)

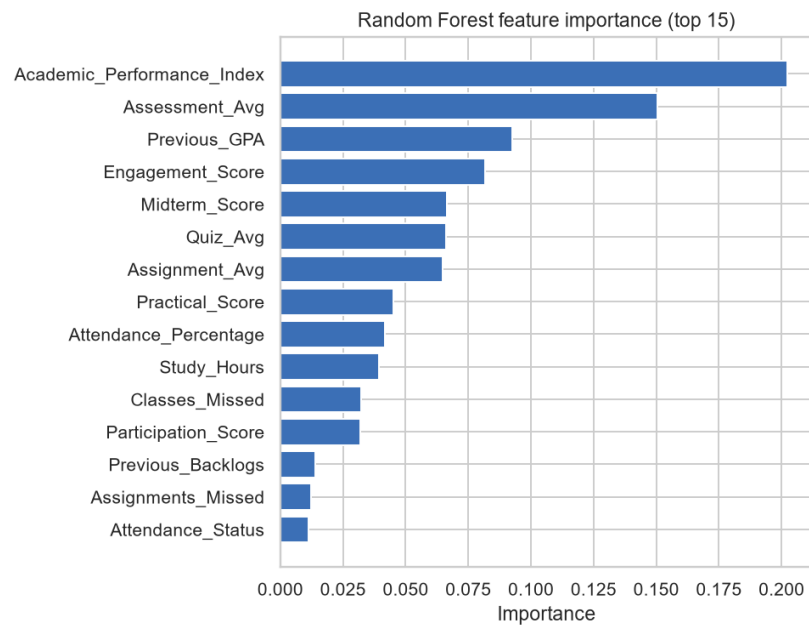
Model	Accuracy	Precision (macro)	Recall (macro)	F1 (macro)	High-risk Precision	High-risk Recall	High-risk F1	ROC-AUC (OvR)
Logistic Regression (baseline)	0.7558	0.7239	0.7509	0.7344	0.6721	0.8098	0.7345	0.9082
Decision Tree	0.735	0.7116	0.73	0.7171	0.6724	0.761	0.714	0.8914
Random Forest (primary)	0.7633	0.741	0.7448	0.7424	0.7391	0.7463	0.7427	0.911
XGBoost	0.75	0.7271	0.7385	0.7315	0.7156	0.761	0.7376	0.9056

For an early-warning system the costly error is a truly high-risk student predicted as Low, so high-risk recall is reported alongside precision and F1. Note that simpler models can achieve a higher high-risk recall than the Random Forest; the decision threshold can be lowered to trade precision for recall depending on how much follow-up capacity the institution has.



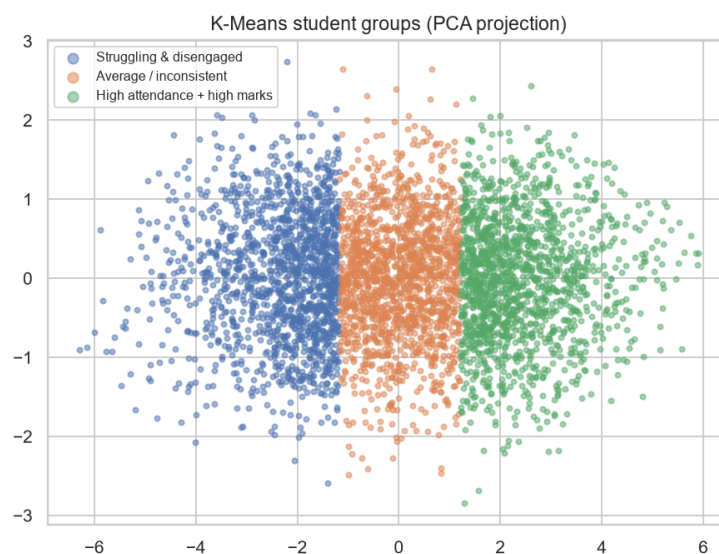
5. Main drivers of risk

feature	importance
Academic_Performance_Index	0.2023
Assessment_Avg	0.1503
Previous_GPA	0.0927
Engagement_Score	0.0818
Midterm_Score	0.0663
Quiz_Avg	0.0662



6. Data mining insights

Cluster_Name	Low	Medium	High
Struggling & disengaged	0.048	0.416	0.535
Average / inconsistent	0.545	0.409	0.046
High attendance + high marks	0.973	0.026	0.001



7. Limitations and responsible use

Results come from synthetic data, so absolute scores do not transfer to a real institution; retrain on real, consented, anonymised data before any use. Predictions are probabilistic support signals, not judgements of a student's ability, and should be reviewed by faculty. Check for group-level bias (e.g. by gender or department) before deployment.